

JARRETT D. PHILLIPS, Ph.D.
School of Computer Science
Department of Integrative Biology
University of Guelph
jphill01@uoguelph.ca
 [Jarrett Phillips](#)
 [jphill01](#)

RESEARCH AREAS

Computational statistics · Machine learning · Data science · Statistical modelling · Biodiversity informatics · eDNA and DNA barcoding

ACADEMIC APPOINTMENTS

Adjunct Professor , School of Computer Science University of Guelph	2023–Present
Affiliated Faculty , One Health Institute University of Guelph	2026

EDUCATION

Ph.D. in Computational Sciences , University of Guelph Co-Advisors: Dr. Daniel Gillis and Dr. Robert Hanner Advisory Committee Members: Dr. Deborah Stacey and Dr. Graham Taylor Dissertation: A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation	2022
Master of Bioinformatics , University of Guelph Co-Advisors: Dr. Robert Hanner and Dr. Daniel Ashlock Major Paper: Assessing DNA Barcode Haplotype Sampling Diversity in the Ray-finned Fishes (Chordata: Actinopterygii)	2014
BSc. (Hons.) in Biological Science, University of Guelph Coursework in bioinformatics, ecology, evolutionary biology, comparative animal physiology, genetics, mathematics, and statistics	2013

RESEARCH EXPERIENCE

Postdoctoral Fellow	2023–Present
University of Guelph	
GBADs Informatics Team	
Stacey Lab, School of Computer Science	
Supervisor: Dr. Deborah Stacey	
Postdoctoral Fellow	2023–Present
University of Guelph	
Gillis Lab, School of Computer Science	
Hanner Lab, Department of Integrative Biology	
Supervisors: Drs. Daniel Gillis and Robert Hanner	
Postdoctoral Fellow	2022
University of Guelph	
Hanner Lab, Department of Integrative Biology	
Supervisor: Dr. Robert Hanner	

RESEARCH FUNDING

Completed

6. Food From Thought Advancing Research Impact (ARIF) Fund – Livestock Innovation Grant	2024
University of Guelph	\$40000.00 CAD
A Modular Decision- and Policy-Making Tool for Global Livestock Disease Burden Assessment	
Role: Co-applicant	
5. Food From Thought Advancing Research Impact (ARIF) Fund	2023
University of Guelph	\$30000.00 CAD
Forecasting Seafood Fraud Occurrence in the Supply Chain Within Major Canadian Cities Using Bayesian Hierarchical Binary Logistic Time-Series Regression Modeling	
Role: HQP	
4. SoCS Travel Grant	2019
University of Guelph	\$1000.00 CAD
3. Arthur D. Latornell Graduate Travel Grant	2019
University of Guelph	\$500.00 CAD
2. CPES Graduate Dean’s Scholarship	2017
University of Guelph	\$3500.00 CAD
1. CPES Graduate Excellence Entrance (GEE) Scholarship	2016
University of Guelph	\$30000.00 CAD

To be Submitted

4. NSERC Discovery Grant	2026
University of Guelph	Amount \$TBD CAD
Computational Statistical Methods for Specimen Sampling Sufficiency and Species Genetic Diversity Assessment	
Role: Principal Investigator	
3. Advancing Research Impact Fund: “Quick Wins” Knowledge Mobilization Fund	2026
University of Guelph	\$20000 CAD

GBADs ModelBuilder as a Tool to Teach Livestock Population Dynamics and Disease Burden Risk Estimation

Role: Co-applicant

2. NSERC Alliance Society Grant

2026

University of Guelph

\$2.5 million CAD

AI-Driven Biodiversity Restoration: Scalable Solutions for Ecosystem Monitoring and Conservation

Role: Collaborator

1. NSERC CREATE Grant

2026

University of Guelph

\$9.9 million CAD

AI-Driven Training of Next-Generation Conservation Scientists

Role: Collaborator

SOFTWARE AND COMPUTATIONAL TOOLS

95,000+ software downloads through the Comprehensive R Archive Network (CRAN)

6. **Seafood Fraud Modelling App** · Python web application

5. **GBADs ModelBuilder** · R Shiny web application

4. **RulesTools** · R package · >3000 downloads · DOI: [10.32614/CRAN.package.RulesTools](https://doi.org/10.32614/CRAN.package.RulesTools)

3. **Frequentist/Bayesian DNA barcode gap coalescent-informed estimators** · R scripts

2. **VLF (Very Low Frequency)** · R package · >40000 downloads · DOI: [10.32614/CRAN.package.VLF](https://doi.org/10.32614/CRAN.package.VLF)

1. **HACSim (Haplotype Accumulation Curve Simulator)** · R package · R Shiny web application ·

>50000 downloads · DOI: [10.32614/CRAN.package.HACSim](https://doi.org/10.32614/CRAN.package.HACSim)

STUDENT SUPERVISION AND MENTORSHIP

Mentored/co-supervised **18 undergraduate and graduate students** across computer science, bioinformatics, statistics, and biology.

* Indicates students under my direct mentorship/supervision

** Indicates students under my co-mentorship/co-supervision

Undergraduate

***Brendan Carl Rosario**

2025

Undergraduate Student Volunteer · Supervised Machine Learning for eDNA spatiotemporal sampling

***Richard Cui**

2025

Summer Undergraduate Research Assistant (URA) · Dynamical modelling of seafood fraud in the supply chain

***Nikolett Toth**

2024-2025

CIS*4900/4910 · Mining association rules for eDNA spatiotemporal sampling

***Nikolett Toth**

2024

Summer Undergraduate Research Assistant (URA) · Mining association rules for eDNA spatiotemporal sampling	
*Fynn De Vuono-Fraser	2024
CIS*4900/4910 · Bayesian modelling of seafood fraud in the Canadian supply chain	
*Zaid Al-Gayyali	2023
Summer Undergraduate Research Assistant (URA) · Seafood Fraud Visualization Tool Shiny app	
*Fynn De Vuono-Fraser	2023
STAT*4600 · Bayesian modelling of seafood fraud in the Canadian supply chain	
*Navdeep Singh	2021
CIS*4900 · HACSim R Shiny web application	
*Scarlett Bootsma	2020–2021
CIS*4900/4910 · HACSim simulation study	
**Maya Persram	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
**Ashley Chen	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
**Olivia Friesen Kroeker	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
*Steven French	2018
CIS*4900/4910 · HACSim R package	
**Julia Harvie	2018–2019
MCB*4500/4510 · Data mining GenBank and BOLD	

Graduate

**Nathan Zeinstra	2024
IBIO*6070 · Bayesian habitat occupancy modelling of sea lamprey using eDNA	
**Amina Asif	2022
BINF*6999 · DNA barcode gap analysis of Canadian disease vectors and agricultural pests	
**Danielle St. Jean (withdrawn)	2018–2019
MSc. thesis (Mathematics) · DNA barcode sequence classification with machine learning	
**Christina Fragerl	2018–2019
BINF*6999 · DNA barcode sequence classification with machine learning	
**Jiaojia (Paula) Yu	2018–2019
BINF*6999 · MDMAPR R Shiny app	
**Ankita Bhandari	2018
BINF*6999 · Data mining GenBank and BOLD	

TEACHING EXPERIENCE

Course Instructor	2025
University of Guelph	
CIS*1910 – Discrete Structures in Computing I	
Graduate Teaching Assistant (GTA)	2016–2020
University of Guelph	

CIS*3130 – System Modelling and Simulation	2020
CIS*1910 – Discrete Structures in Computing I	2017
CIS*2460 – Modelling of Computer Systems	2016–2019

ACADEMIC SERVICE

Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference University of Guelph	2026
Conference organizer and volunteer	
39th Canadian Artificial Intelligence Conference Simon Fraser University	2026
Reviewer for Long and Short Papers Track	
GBADs Informatics Technical Workshop University of Liverpool	2023
Workshop organizer	
Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference University of Guelph	2023
Conference organizer and volunteer	
School of Computer Science (SoCS) Faculty Search Committee University of Guelph	2018
Member of hiring panel for Associate Professor in Cybersecurity	
School of Computer Science (SoCS) Faculty Search Committee University of Guelph	2017–2018
Member of hiring panel for 2-year Contractually Limited Assistant Professor in Cybersecurity	

CONFERENCE PRESENTATION AND PROCEEDINGS

* Indicates students under my direct mentorship/supervision

16. The VII AMMCS International Conference (student oral presentation) Wilfrid Laurier University, Canada · Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain	2026
15. Pathway to Increase Standards and Competency of eDNA Surveys (PISCeS) International Conference (oral presentation) University of Guelph, Canada	2026
Phillips, J.D. · Association rule mining for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling	
14. Pathway to Increase Standards and Competency of eDNA Surveys (PISCeS) International Conference (poster presentation) University of Guelph, Canada	2026
*Toth, N., Antonie, M.L. and Phillips, J.D. · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets	
13. 39th Canadian Conference on Artificial Intelligence (short paper and student poster) Simon Fraser University, Canada	2026

- *Toth, N., Antonie, M.L. and **Phillips, J.D.** · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets.
12. **SoCS Undergraduate Summer Project Show and Tell** (student oral presentation) 2025
University of Guelph, Canada
- *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** · The Importance in Design as a Computer Scientist
11. **CEPS Undergraduate Student Poster Day** (student poster presentation) 2024
University of Guelph, Canada
- *Toth N. and **Phillips, J.D.** · Association Rule Mining of eDNA Datasets
10. **CBS Undergraduate Poster Session** (student poster presentation) 2024
University of Guelph, Canada
- *Toth N. and **Phillips, J.D.** · Association Rule Mining of eDNA Datasets
9. **9th International Barcode of Life Conference** (poster presentation) 2024
Estação das Docas, Belem, Brazil
- Phillips, J.D.** · A Measure of the DNA Barcode Gap for Applied and Basic Research (*not attended*)
8. **GBADs Informatics Technical Workshop** (oral presentation) 2023
University of Liverpool, England
- Phillips, J.D.** · The GBADs R Package (and Why We Need It!)
7. **CEPS Student Research Day** (student poster presentation) 2023
University of Guelph, Canada
- *De Vuono-Fraser, F.A. and **Phillips, J.D.** · Estimating Seafood Mislabelling Rates in Canada Using Bayesian Modelling
6. **Pathway to Increase Standards and Competency of eDNA Surveys (PISCes)** 2023
International Conference (poster presentation)
University of Guelph, Canada
- Phillips, J.D.** · Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes
5. **8th International Barcode of Life Conference** (oral presentation) 2019
NTNU University Museum and Norwegian Biodiversity Information Centre, Norway
- Phillips, J.D.** · HACSIm: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness
4. **Guelph BioMathematics and Statistics (BioM&S) Symposium**
Artificial Intelligence and Machine Learning in Biology (attended) 2019
University of Guelph, Canada
3. **CEPS Undergraduate Student Poster Day** (student poster presentation) 2018
University of Guelph, Canada
- *French, S.H. and **Phillips, J.D.** · Estimating Sampling Size Using Haplotype Accumulation Curves and Semiparametric Models
2. **7th International Barcode of Life Conference** (oral presentation). 2017
University of Johannesburg, South Africa
- Phillips, J.D.** · Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough?
1. **6th International Barcode of Life Conference** (poster presentation) 2015
University of Guelph, Canada

Phillips, J.D. · An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii).

REFEREED WORK

Publication record: 8 peer-reviewed journal articles (6 first-author, 2 co-author) and 2 book chapters (1 first-author, 1 co-author), plus 3 first-author manuscripts in preprint or under review. Senior (last) author on 2 student-led manuscripts

* Indicates students under my direct mentorship/supervision

** Indicates students under my co-mentorship/co-supervision

Journal Articles

Published or Accepted

9. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2026) Swimming in uncertainty: How proper statistical modelling can help expose seafood product mislabeling. *CHANCE*, **3**. DOI: 10.1080/09332480.2026.2725526
8. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2026). Statistical modelling of seafood fraud highlights uncertainties in products from Metro Vancouver, British Columbia, Canada: Revisiting Hu *et al.* (2018). *Journal of Food Science*, **91**: e71201. DOI: [10.1111/1750-3841.71201](https://doi.org/10.1111/1750-3841.71201)
7. Raymond, K., Sobkowich, K.E., **Phillips, J.D.**, Nguyen, L., McKechnie, I., Mohideen, R.N., Fitzjohn, W., Szurkowski, M., Davidson, J., Rushton, J., Stacey, D.A., and Bernardo T.M. (2024). GBADs informatics strategy: User-centric tools, data quality, and model interoperability. *WOAH Scientific and Technical Review*, **43**: 96-107. DOI: [10.20506/rst.43.3522](https://doi.org/10.20506/rst.43.3522).
6. **Phillips, J.D.**, Athey, T.B.T., Hanner, R.H., and McNicholas, P.D. (2023). VLF: An R package for the analysis of very low frequency variants in DNA sequences. *Biodiversity Data Journal*, e96480. DOI: [10.3897/BDJ.11.e96480](https://doi.org/10.3897/BDJ.11.e96480).
5. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2022). Lack of statistical rigor in DNA barcoding likely invalidates the presence of a true species' barcode gap. *Frontiers in Ecology and Evolution*, **10**: 859099. DOI: [10.3389/fevo.2022.859099](https://doi.org/10.3389/fevo.2022.859099).
4. D'Ercole, J., Dincă, V., Opler, P.A., Kondla, N.G., Schmidt, C.B., **Phillips, J.D.**, Robbins, R., Burns, J.M., Miller, S.E., Grishin, N., Zakharov, E.V., deWaard, J.R., Ratnasingham, S., and Hebert, P.D.N. (2021). A DNA barcode library for the butterflies of North America. *PeerJ*, **9**: e11157. DOI: [10.7717/peerj.11157](https://doi.org/10.7717/peerj.11157).
3. **Phillips, J.D.**, *French, S.H., Hanner, R.H., and Gillis, D.J. (2020). HACSim: An R package to estimate intraspecific sample sizes for genetic diversity assessment using haplotype accumulation curves. *PeerJ Computer Science*, **6**(192): 1-37. DOI: [10.7717/peerj-cs.243](https://doi.org/10.7717/peerj-cs.243).
2. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2019). Incomplete estimates of genetic diversity within species: Implications for DNA barcoding. *Ecology and Evolution*, **9**(5): 2996-3010. DOI: [10.1002/ece3.4757](https://doi.org/10.1002/ece3.4757).
1. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii). *DNA Barcodes*, **3**: 66-73. DOI: [10.1515/dna-2015-0008](https://doi.org/10.1515/dna-2015-0008).

Preprints

3. *Toth, N., Antonie, L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. bioRxiv. DOI: [10.64898/2026.09.16.752056](https://doi.org/10.64898/2026.09.16.752056).
2. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. (2025). A Bayesian coalescent model of the DNA barcode gap. Authorea. DOI: [10.22541/au.174073683.37707806/v1](https://doi.org/10.22541/au.174073683.37707806/v1)
1. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2024). Statistical modelling of seafood fraud in the Canadian supply chain. bioRxiv. DOI: [10.1101/2024.02.05.578947](https://doi.org/10.1101/2024.02.05.578947).

Submitted or Under Revision

1. *Toth, N., Antonie, L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. Submitted to *Oikos*.

To Be Submitted

2. Cui, R.C., Yodzis, M.P., and Phillips, J.D. Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain. Springer Nature.
1. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. A Bayesian-informed coalescent model of the DNA barcode gap. Targeted to *Methods in Ecology and Evolution*.

In preparation (selected)

3. *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** Seafood market product fraud erodes both consumer trust and resource sustainability within the global supply chain. Targeted to *Proceedings of the Royal Society B: Biological Sciences*.
2. Elliott, T.A., Diao, J., **Phillips, J.D.**, and Adamowicz, S.J. Benchmarking imputation methods for insect trait data across missingness mechanisms and trait types. Targeted to *Methods in Ecology and Evolution*.
1. D'Ercole, J., Dapporto, L., **Phillips, J.D.**, Dincă, V.E., Vila, R., Talavera, G., and Hebert, P.D.N. Macrogenetics of North American butterflies—The impact of Quaternary climatic fluctuations. Targeted to *PNAS*.

Book Chapters

Published or Accepted

2. **Phillips, J.D.**, Griswold, C.K., Young, R.G., Hubert, N., and Hanner, R.H. (2024). A Measure of the DNA Barcode Gap for Applied and Basic Research. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY.
URL: https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_24
1. Hubert, N., **Phillips, J.D.**, and Hanner, R.H. (2024). Delimiting Species with Single-Locus DNA Sequences. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY. URL: https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_3

Submitted or Under Revision

1. *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain. VII AMMCS Conference, Springer Nature.
- #### **Conference Proceedings**

Published or Accepted

5. *Toth, N., Antonie, M.L., and **Phillips, J.D.** To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets. *39th Canadian Conference on Artificial Intelligence*, **318**: 1012-1019. Short paper and poster.
4. Morey, K., Loeza-Quintana, T., **Phillips, J.**, and Hanner, R. (2023). Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes. Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference. Poster.
3. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2019). HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness. Scientific abstracts from the 8th International Barcode of Life Conference, Trondheim, Norway (ed. Torbjørn Ekrem), *Genome*, 62(6): 349-453. Oral presentation.
2. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2017). Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough? Scientific abstracts from the 7th International Barcode of Life Conference, Johannesburg, South Africa (ed. M. van der Bank), *Genome*, 60(11): 881-1019. Oral presentation.
1. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific haplotype diversity in the ray-finned fishes (Chordata: Actinopterygii). Scientific abstracts from the 6th International Barcode of Life Conference, Guelph, ON., Canada (ed. S.J. Adamowicz), *Genome*, 58(5): 163-303. Poster.

NON-REFEREED WORK

Ph.D. Thesis

Phillips, J.D. (2022). A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation. University of Guelph Atrium. URL: <https://atrium.lib.uoguelph.ca/items/8addfcc5-f21c-4691-89b7-c4db051892eb>

* Indicates students under my direct mentorship/supervision

15. *Cui, R.C. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House
14. *Toth, N. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House
13. *Cui, R.C. and **Phillips, J.D.** (2025). The Importance in Design as a Computer Scientist. Student-contributed SoCS Show and Tell video URL: <https://github.com/RichaCui70/Dynamics-between-Seafood-Fraudsters-and-Buyers/blob/main/Summer%20Showcase%20-%20Richard%20Cui.mp4>
12. **Phillips, J.D.**, *De Vuono-Fraser, F.A., Gillis, D.J., and Hanner, R.H. (2024). Statistical modelling of seafood fraud. Whiteboard explainer video. URL: <https://www.youtube.com/watch?v=4jU8cU8M70U>
11. *Toth, N. and **Phillips, J.D.** (2024). eDNA Collection Gets a Tech Update. Student-contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2024/07/edna-collection-gets-tech-upgrade>
10. *Toth, N. and **Phillips, J.D.** (2024). Unravelling eDNA with Association Rule Mining. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2024/08/15/unravelling-edna-with-association-rule-mining/>

9. **Phillips, J.D.** (2024). Summer URA Position. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2024/02/07/summer-ura-position-2/>
8. **Phillips, J.D.** (2022). A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation. Ph.D. Thesis. URL: <https://atrium.lib.uoguelph.ca/items/8addfcc5-f21c-4691-89b7-c4db051892eb>
7. **Phillips, J.D.** (2022). Mind the Gap — The DNA Barcode Gap, That Is. Contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2022/08/mind-gap---dna-barcode-gap>
6. **Phillips, J.D.** (2020). Barcode Cracking. Contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2020/02/barcode-cracking>
5. **Phillips, J.D.** (2020). Protecting Biodiversity Through the Lens of Genetic Diversity. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2020/01/30/protecting-biodiversity-through-the-lens-of-genetic-diversity/>
4. **Phillips, J.D.** (2019). IBOL8 and the Midnight Sun. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2019/07/02/reflections-ibol8-and-the-midnight-sun/>
3. **Phillips, J.D.** (2017). The Big Five and IBOL7. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2017/12/06/reflections-the-big-five-and-ibol7/>
2. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding: Are current sampling levels enough? Contributed guest post to the DNA Barcoding Blog of Dr. Dirk Steinke. URL: <http://dna-barcoding.blogspot.com/2016/01/guest-post-sample-size-estimation-for.html>
1. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding of ray-finned fishes: Are current sampling levels enough? Contributed newsletter article to the Barcode Bulletin, 7(1). URL: <https://journal.lib.uoguelph.ca/index.php/ibol/issue/view/345>

ACADEMIC PEER REVIEW SERVICE

Ecology and Evolution, F1000 Research, Frontiers in Ecology and Evolution, Lifestyle Genomics, Mitochondrial DNA Part A, Molecular Ecology Resources, Molecular Biology Reports, Methods in Ecology and Evolution, Nature Communications

VOLUNTEER EXPERIENCE

- | | |
|--|------------------------------|
| <p>CEPS Student Research Connections Networking Night
University of Guelph</p> <ul style="list-style-type: none"> • Connected with undergraduate students for summer URA project recruitment <p>CIS*3750 Wireframing session
University of Guelph</p> <ul style="list-style-type: none"> • Graded students on mobile app prototypes for various community partners using Qualtrics surveys | <p>2025</p> <p>2021-2025</p> |
|--|------------------------------|